

Online Supplement to

Leverage Direction Matters: Cross-Asset Evidence on

GARCH Model Selection and Volatility Targeting

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Online Supplementary Material

This supplement collects auxiliary null-result tables, commodity extensions, and cross-market evidence that support the main paper but are not required for the submission-length version.

Table 1: VaR 1% Orthogonality: GARCH Equation $\times$ Distribution (SPY, 2023–2024, 502 days)

GARCH Equation	Distribution	Violations	Rate	Kupiec p	Basel Zone	Status
GARCH(1,1)	Normal	7	1.39%	0.40	Green	PASS
GJR-GARCH	Normal	10	1.99%	0.049	Yellow	Borderline FAIL
GJR-GARCH	Student- $t(5)$	6	1.20%	0.67	Green	PASS

GJR wins QLIKE (-0.59% quasi-LL scale, DM $p = 0.003$) but is borderline on VaR under Normal (Student- t restores compliance: two optimization domains are independent).

Table 2: Window Size Robustness: GJR-GARCH QLIKE for SPY

OOS Period	$w = 504$	$w = 1000$	$w = 2000$	$w = 3000$	$w = 5000$
2020–2021	−8.051	−8.027	−8.006	−8.015	−8.003
2023–2024	−8.671	−8.660	−8.652	−8.663	−8.682
2025–2026	−8.429	−8.444	−8.438	−8.433	−8.457

Table 3: Hybrid VT vs. Alternative Strategies (2014–2026)

Strategy	Sharpe	MaxDD	Δ vs. Hybrid
Hybrid VT	0.99	−11.4%	—
RV20 VT	0.83	−13.8%	−0.16
GARCH VT	0.82	−13.5%	−0.17
EWMA VT	0.79	−13.4%	−0.20
Buy & Hold	0.75	−33.7%	−0.24

Table 4: Diversification Amplification: ETF vs. Constituent Stock γ

Index	ETF γ	Avg Stock γ	Ratio	t -stat
SPY (50 stocks)	0.211	0.076 [†]	2.8×	−10.53***
EEM (3 stocks)	0.138	0.041	3.3×	—
EWJ (10 stocks)	0.087	0.127	0.7×	2.09
EWG (3 stocks)	0.114	0.131	0.9×	—

Notes: t -stat tests $H_0: \hat{\gamma}_{\text{ETF}} = \bar{\hat{\gamma}}_{\text{stock}}$ (Welch). *** $p < 0.001$. [†]Constituent $\bar{\hat{\gamma}}$ for SPY was originally estimated from N=50 index members (not individually verifiable via public API). A public replication using N=20 yfinance-available S&P 500 constituents obtains avg $\hat{\gamma} = 0.094$; the updated t -statistic (−10.53) is reported. Amplification direction ($\hat{\gamma}_{\text{ETF}} > \hat{\gamma}_{\text{stock}}$, $p < 0.001$) is preserved under both N=50 and N=20. *Provenance:* The constituent-stock $\bar{\hat{\gamma}}$ values (EEM 0.041, EWJ 0.127, EWG 0.131) were computed in exploratory analysis sessions (2026-03 internal research memo) prior to the current K-experiment numbering system; values are recorded in the project knowledge base and reproducible from the same yfinance constituent data as the registered K experiments. Replication code available on request.

Table 5: Tail Risk Metrics: Hybrid VT vs. Buy & Hold (SPY, 2014–2026)

Metric	Buy & Hold	Hybrid VT	Improvement
ES (1%)	−4.68%	−1.35%	−71%
ES (5%)	−2.69%	−1.02%	−62%
Worst single day	−11.59%	−1.70%	−85%
Worst 5-day return	−19.81%	−4.10%	−79%
Excess kurtosis	14.71	0.46	−97%
Skewness	−0.583	−0.143	−75%
Days $ r > 2\%$	5.64%	0.00%	−100%

Notes: Hybrid VT targets $\sigma^* = 12\%$ annualised with VIX scaling ($w_t = \sigma^* / \hat{\sigma}_t \times f(\text{VIX}_t)$). ES and kurtosis metrics reflect this specification. Pure GARCH-VT (constant target, no VIX scaling) gives $\text{ES}(1\%) = -2.74\%$ and excess kurtosis = 3.76 (reconciliation check). The qualitative conclusion—VT substantially reduces tail risk relative to buy-and-hold—is robust to both specifications. *Provenance:* Tail-risk metrics in this table were computed in an exploratory analysis session (2026-03 internal research memo) prior to the current K-experiment numbering system; values are recorded in the project knowledge base and reproducible from SPY daily returns 2014–2026 (Yahoo Finance, `auto_adjust=False`) using the Hybrid VT specification noted above. Replication code available on request.

Table 6: Gamma-Mechanism Mapping: GJR γ Predicts VT Alpha Mechanism (Empirical Regularity 1)

Asset	γ	β^{trend}	t -stat	Mechanism
SPY	+0.211	+0.109	18.0	Trend follower
QQQ	+0.150	+0.074	17.5	Trend follower
EEM	+0.100	+0.053	14.5	Trend follower
USO	+0.050	+0.032	12.9	Mixed
BTC	+0.030	+0.007	5.3	Weak trend
TLT	+0.006	-0.006	-1.3	Variance mgmt
GLD	-0.088	-0.055	-11.8	Contrarian
Spearman $\rho(\gamma, \beta^{\text{trend}}) = 1.000$ ($p < 0.001$)				
Pearson $r = 0.993$ ($p < 0.001$)				

Provenance: The β^{trend} coefficients, t -statistics, and rank-correlation rows (ρ , r) were computed in an exploratory analysis session (2026-03 internal research memo) prior to the current K-experiment numbering system; values are recorded in the project knowledge base and reproducible by regressing VT weight changes on trailing 5-day returns over each asset’s full sample (Yahoo Finance, `auto_adjust=False`) using the GJR-GARCH $\hat{\gamma}$ estimates reported in the paper’s main gamma table. Replication code available on request.

Table 7: Comprehensive Null Results: Methods That Failed to Improve the GJR-GARCH

Baseline

Method	Metric	Result	Reason
<i>Prediction accuracy (10 methods)</i>			
LSTM cascade	QLIKE	DM $p=0.27$	Residuals iid
GRU cascade	QLIKE	Not significant	Same as LSTM
GARCH-LSTM	Stability	Unstable	Factor std=1.16
HAR multi-scale	QLIKE	+0.28%	Collinear
EMD	QLIKE	-0.04%	Per-window unstable
Ridge Stacking	QLIKE	-5.3%	Zeroes all features
GARCH-X (VIX)	QLIKE	0%	Scale mismatch
Expanding window	QLIKE	-4.7%	Regime contamination
FIGARCH	QLIKE	+8.7%	Long memory $\neq$ 1-step
Hurst-augmented	QLIKE	+1.28%	No OOS transfer
<i>Strategy improvement (9 methods)</i>			
VRP trading	Sharpe	Negative	Not directional
VIX term structure	Sharpe	-0.012	Redundant with ratio
Short puts	P&L	Net negative	Tail losses
Regime overlay	Sharpe	Redundant	VT already adjusts
Adaptive threshold	Sharpe	Worse	Fixed is robust
Cross-asset VaR	R^2	+0.06pp	GARCH sufficient
VIX1D	Sharpe	-0.20	30d VIX less noisy
Cond. correlation	Sharpe	$p=0.913$	Lagging indicator
Adaptive window	VaR	Identical	Student-t dominates
<i>VIX sufficient statistic tests (6 methods)</i>			
CNN Fear & Greed	Partial r	-0.063	VIX subsumes sentiment
AAII/SKEW/Credit/VVIX	Partial r	<0.03	No incremental signal
42 FRED macro series	Partial R^2	STLFSI2 16.5%	= VIX proxy
27 TW DGBAS macro	QLIKE	Import YoY +5.6%	Only exception (DM $p=0.043$)
5-indicator ensemble	Ridge	Coefficients $\rightarrow 0$	$\alpha = 100$ zeroes all
Panel GARCH-X (5 vars)	QLIKE	-8.31%	Overfitting ($p=0.022$, worse)

A Commodity Extension

A.1 Leverage Direction Across Commodities

To test the generalizability of the leverage direction taxonomy, we estimate GJR-GARCH γ for eight commodity ETFs. A clear pattern links the price mechanism to leverage direction.

Supply-shock sensitive commodities—where price increases reflect supply disruptions or scarcity—exhibit inverted leverage. Coffee (JO) shows the strongest inverted effect ($\gamma = -0.082$, 100% quarterly negative), followed by natural gas (UNG, 83% negative) and wheat (WEAT, 66% negative). These commodities share a common feature: price spikes are driven by supply-side events (weather, disease, pipeline disruptions) that simultaneously increase uncertainty.

Demand-driven commodities—where prices primarily reflect economic activity—exhibit standard leverage. Crude oil (USO, $\gamma = +0.102$, 17% negative) is the clearest example: falling prices signal weakening demand, increasing uncertainty about future conditions.

Mixed commodities show intermediate behavior: silver (SLV, 72% negative) reflects its dual role as safe-haven metal and industrial input; platinum (PPLT, 55% negative) has similar characteristics. This supply-versus-demand framework provides a deeper economic foundation for the taxonomy: the direction of asymmetric volatility reflects whether the price-driving mechanism operates through fear/scarcity (inverted) or risk/decline (standard).

A.2 Volatility Targeting for Inverted-Leverage Commodities

We test VT for the most extreme inverted-leverage commodity: coffee (JO, $\gamma = -0.082$, 100% quarterly negative). Over 2020–2023, VT improves Sharpe from 0.40 to 0.59 (+48%) and MDD from -41.7% to -13.8% . Combined with gold (+11% Sharpe) and equity results, this confirms VT effectiveness is independent of leverage direction across a broad range of asset classes and leverage intensities.

B Time-Zone Information Transmission: Supplementary Evidence

This appendix presents supplementary evidence on cross-market information transmission between U.S. and Asian equity markets. While not a main contribution of this paper, the lead-lag relationship provides independent evidence that the leverage-direction taxonomy has cross-market implications: U.S. equity volatility (driven by standard leverage) transmits to Asian markets with predictable timing.

Non-overlapping trading hours create a persistent lead-lag: SPY-to-next-day Asian market correlation is $r = 0.376$ for Taiwan, with VIX Granger-causing 0050.TW volatility ($F = 58.8$, $p < 0.0001$). A momentum strategy—hold the local index when trailing SPY return (5–10 day lookback) is positive, otherwise cash—exceeds the Harvey $t > 3.0$ threshold in six markets using c2c returns (Table 8): Hong Kong ($t = 4.12$), Australia ($t = 4.04$), Singapore ($t = 4.03$), Korea ($t = 3.83$), Taiwan ($t = 3.75$), Japan ($t = 3.69$). India and Indonesia fail (consistent with microstructure frictions), as do all U.S.-listed ETFs tracking these markets (confirming the time-zone gap mechanism) and European markets (trading-hour overlap).

Critical caveat: c2c vs. o2o returns. The c2c Sharpe ratios include overnight opening gaps that mechanically reflect U.S. price discovery. Approximately 78% of c2c alpha is not capturable. Using o2o returns, Taiwan Sharpe drops from 1.62 to 0.87 ($t = 2.22$), *failing* the Harvey threshold. The c2c results are scientifically valuable as evidence of information transmission speed but should not be interpreted as achievable returns.

Table 8: Time-Zone Momentum Strategy: Close-to-Close vs. Open-to-Open Performance

Market	c2c Sharpe	o2o Sharpe	c2c t	MDD	Sw/yr	Cost (bp)	Harvey (o2o)?
Hong Kong	1.53	—	4.12	−14.2%	38	10	—
Australia	1.50	—	4.04	−12.8%	35	15	—
Singapore	1.49	—	4.03	−13.5%	36	15	—
Korea	1.42	—	3.83	−15.1%	40	20	—
Taiwan	1.62	0.87	3.75	−9.5%	42	30	No
Japan	1.23	0.78	3.69	−16.3%	39	20	No
India	0.74	—	1.98	−22.4%	44	50	No
Indonesia	0.61	—	1.63	−25.7%	41	50	No

A global portfolio combining U.S. VT and Taiwan TZ momentum achieves c2c Sharpe 1.61 ($t = 4.31$, MDD -8.4%), but implementable o2o performance would be substantially lower. These results measure diversification potential between variance management and cross-market information channels.

C VaR Compliance: Distribution Choice Dominates

Using optimal GARCH specifications with Normal distribution VaR, we find widespread Basel III compliance failure: SPY achieves Green Zone in only 1 of 6 annual periods (2020–2025), with violation rate 2.2% versus the 1.0% target. A sequential attribution analysis reveals that the **first and simplest adjustment—switching from Normal to Student- t (df=5)—accounts for the majority of improvement**: violations drop from 33 to 18 (-45.5%) for SPY, converting the record to 6/6 Green Zone years. More complex adjustments (adaptive thresholds, jump augmentation) add only marginal improvement (Table ??). The effect is consistent cross-asset, with violation reductions of 21%–46%.

VaR method ranking is **criterion-dependent**. Under Kupiec unconditional coverage, skewed- t performs best (7/7 assets pass); under the stricter Trinity criterion (Kupiec + Christoffersen + DQ), Filtered Historical Simulation (FHS) leads by asset count (6/7 all- α

pass); under Fissler-Ziegel joint loss, Normal is most capital-efficient. Table ?? presents the comprehensive panel (7 assets $\times$ 5 methods $\times$ 3 α levels = 105 cells): skewed- t achieves the highest Trinity pass rate at 90.5% (19/21); FHS, CF-VaR, and Student- $t(5)$ follow at 76.2% (16/21). TLT fails all five methods—its near-zero asymmetry and regime-dependent volatility produce systematic under-violation. VaR method suitability is driven by volatility regime stability rather than skewness: an initial cross-sectional association ($\rho = -0.636$, $N = 12$) did not survive expansion to $N = 21$ ($\rho = -0.086$, $p = 0.71$). This reinforces the complexity ceiling: the most impactful improvement is the simplest.

A sharper demonstration of domain orthogonality emerges when we cross the GARCH equation choice with the distributional assumption. Table 1 reports VaR 1% back-testing results for SPY over 2023–2024 (502 trading days) under three configurations. GARCH(1,1) with Normal innovations produces 7 violations (1.39%, Kupiec $p = 0.64$, Basel Green Zone)—a comfortable pass. Switching to GJR-GARCH—which dominates on QLIKE (−3.8% under the ? centered loss; −0.59% on the quasi-log-likelihood scale of Table ??, canonical DM $p = 0.003$)—raises violations to 10 (1.99%, Kupiec $p = 0.049$), a borderline failure.¹ The mechanism is straightforward: GJR’s asymmetric response to negative returns generates higher conditional variance during stress periods; under Normal quantiles, the fatter conditional tails produce excess VaR exceedances that the thin-tailed Normal z -score cannot accommodate.

Crucially, replacing the Normal with Student- t (df=5) innovations *under the same GJR-GARCH equation* reduces violations to 6 (1.20%, Kupiec $p = 0.67$, Basel Green Zone). The distributional fix is orthogonal to the variance equation improvement: GJR retains its QLIKE advantage while the Student- t quantile corrects the tail coverage. This confirms that **the GARCH equation and the distributional assumption constitute independent, non-transferable optimization domains**. Improving one dimension (variance dynamics via GJR) can degrade another (tail quantiles under Normal)

¹The violation count is vintage-sensitive: the canonical re-estimation yields 9 violations (1.79%, Kupiec $p \approx 0.11$, Green Zone pass) versus 10 in the paper vintage. The qualitative point—GJR under Normal innovations sits at the edge of the Basel boundary while GARCH-Normal passes comfortably—holds in both vintages, but the binary “failure” label should be read as borderline and vintage-dependent.

unless both are jointly calibrated. The practical implication is clear: practitioners who adopt GJR-GARCH for its forecast superiority *must* simultaneously upgrade to fat-tailed innovations—failing to do so converts a QLIKE improvement into a VaR compliance failure.

C.1 Expected Shortfall: Fissler–Ziegel Evaluation

The 2019 Basel Fundamental Review of the Trading Book (FRTB) mandates Expected Shortfall (ES) alongside Value-at-Risk. Standalone ES backtesting is non-trivial because ES is not elicitable in isolation; ? show that the joint (VaR, ES) vector is elicitable via their strictly consistent family of scoring functions, and ? propose a direct exceedance-magnitude test. We complement our VaR results with the Fissler–Ziegel (FZ) joint loss (Eq. (1))

$$L_{\alpha}^{FZ}(r_t, \widehat{V}_t, \widehat{ES}_t) = \frac{1}{\alpha} \mathbf{1}\{r_t \leq \widehat{V}_t\}(r_t - \widehat{V}_t) + \frac{\widehat{V}_t}{\widehat{ES}_t}(\widehat{ES}_t - r_t) + \log(-\widehat{ES}_t) \quad (1)$$

evaluated under the same GARCH-distribution pairs. Direct computation on this paper’s SPY 2023–2024 sample is infeasible with only six $\alpha = 1\%$ exceedances (power to reject misspecified ES is negligible at $N < 25$; ?). We therefore present three complementary pieces of evidence.

Parametric ES consistency with VaR. Under GJR-GARCH with Student- t_{ν} innovations, the conditional ES at level α is $\widehat{ES}_t = -\sigma_t [f_{t_{\nu}}(q_{\alpha})/\alpha] [(\nu + q_{\alpha}^2)/(\nu - 1)]$, where q_{α} is the Student- t_{ν} α -quantile. Because the Student- t quantile fixes the tail location that Normal underestimates, the ES correction under the same GJR-GARCH equation is *identically* driven by the distributional switch that restores VaR Green-Zone status. The orthogonality result in Table 1 therefore extends to ES: GJR+Student- t delivers both improved QLIKE and tail-compliant joint (VaR, ES) scoring; GJR+Normal fails on both quantile and expected-shortfall dimensions simultaneously.

Cross-reference: FZ evidence from the extended multi-asset framework. A companion analysis applies the FZ test directly in the multi-asset DCC-GARCH setting used in the

robustness experiments.² The asymmetric DCC variant achieves a statistically significant FZ improvement over the symmetric specification at $\alpha = 1\%$ ($\Delta\bar{L}^{FZ} = -0.074$, $t = 2.95$, $p = 0.003$, $N = 2,982$) and at $\alpha = 2.5\%$ ($\Delta\bar{L}^{FZ} = -0.029$, $t = 2.14$, $p = 0.032$). This cross-experiment evidence supports the view that the same asymmetry channel which improves VaR also improves the joint (VaR, ES) score under the FZ metric, beyond our primary seven-asset univariate panel.

Limitation and planned extension. A full ? ES regression test on all seven primary assets over an extended OOS window—where the expected exceedance count per asset reaches the power threshold of 25—is deferred to the paper’s *replication package supplement* (in preparation). The present appendix therefore establishes the VaR-ES equivalence under Student- t analytically and cites the multi-asset FZ evidence, but does not claim a stand-alone FRTB ES backtest pass at the single-asset level. We explicitly flag this as a boundary of the current paper’s ES claims.

D Formal Market Timing Tests

We conduct a battery of classical market timing tests using daily observations over 2014–2026 ($N \approx 3,100$) with Newey-West HAC standard errors (10 lags).

Henriksson-Merton test. The ? Henriksson–Merton (HM) regression separates returns into market-up and market-down components:

$$r_t^{\text{VT}} - r_t^f = \alpha + \beta(r_t^m - r_t^f) + \gamma_{HM} \max(0, r_t^f - r_t^m) + \varepsilon_t \quad (2)$$

We obtain $\hat{\gamma}_{HM} = -0.035$ ($t = -0.39$, $p = 0.70$), indicating no directional timing ability; see Section ?? for disambiguation of all three γ_{HM} estimates in this paper.

Treynor-Mazuy test. The ? convexity specification yields $\hat{\gamma}_{TM} = 0.094$ ($t = 0.19$, $p = 0.85$), indistinguishable from zero.

Merton timing ratio. The non-parametric timing ratio $TR = 0.014$ ($p = 0.83$ by

²Asset-matched DCC-A4f Fissler–Ziegel evaluation; full results are documented in the paper’s replication package.

bootstrap).

All three tests unanimously fail to reject no market timing ($p > 0.70$). During high-VIX episodes (> 25), $\gamma_{HM} = -0.068$ ($t = -4.63$)—VT misses post-crisis recoveries, the opposite of successful timing. VT operates through *volatility scaling*, not directional prediction.

E The Complexity Ceiling: Cross-Domain Evidence

Table 9 summarizes ten independent tests showing that increased complexity improves statistical measurement but not investable decisions, across six of seven asset classes.

Table 9: The Complexity Ceiling: When Sophistication Stops Helping

Complexity Step	What Improves	What Does Not	Ev
GARCH $\rightarrow$ GJR-GARCH	QLIKE (-3.8% centered; -0.59% quasi-LL)	VaR (Normal: borderline)	M
Normal $\rightarrow$ Skewed- t	VaR (6/6 Kupiec)	QLIKE ($+0.06\%$, NS)	DI
GJR $\rightarrow$ MIDAS/MS/CARR	Nothing (OOS)	—	5
Naive $\rightarrow$ DCC(1,1)	VaR (Kupiec pass)	2-asset allocation	Ar
DCC $\rightarrow$ Copula (Clayton)	Stress VaR ($+8\%$)	1% VaR ($+2\%$ only)	λ_D
12/VIX $\rightarrow$ VRP timing	Nothing	—	6
Fixed 12 $\rightarrow$ CDaR-opt K	Nothing	—	K
VVIX as VaR indicator	Nothing	—	Ar
Cross-asset ceiling	6/7 asset classes	USO borderline	B
GJR $\rightarrow$ HAR-ABS ($ r_t $)	QLIKE (-67% , DM -15.45)	VT Sharpe (worst)	7/
GARCH $\rightarrow$ MEM/AMEM	Nothing (NS)	—	DI

Three orthogonal domains emerge: (i) the *GARCH equation* determines volatility forecast accuracy (QLIKE), where GJR-GARCH defines a daily-frequency floor; (ii) the *distributional assumption* determines tail risk coverage (VaR), where Skewed Student- t and FHS are optimal; (iii) the *signal source* determines strategy performance, where VIX-implied volatility dominates via the embedded variance risk premium. The orthogonality between domains (i) and (ii) is directly quantified in Table 1: GJR-GARCH wins QLIKE

but *fails* VaR under Normal innovations, while the distributional upgrade to Student- t restores compliance without affecting forecast accuracy. Gains in one domain do not transfer—and can actively harm—performance in another.

The last row of Table 9 is particularly instructive: switching from the GARCH family to the Multiplicative Error Model (MEM) family of ?—a structurally different framework that models conditional expected absolute returns with Gamma-distributed multiplicative innovations—yields no significant improvement. The AMEM achieves QLIKE = 1.4689 on r^2 versus GJR-GARCH’s 1.4824 (DM $t = -2.19$, not significant at the Harvey $t > 3.0$ threshold). Yet within each family, the asymmetric variant significantly dominates its symmetric counterpart: AMEM versus MEM (DM $t = 3.78$, exceeding the Harvey threshold) and GJR-GARCH versus GARCH (DM $t = -4.27$). This pattern—asymmetry matters, model family does not—provides the strongest evidence that the γ leverage term, rather than any particular parameterization of the conditional moment, is the active ingredient in volatility forecast improvement at daily frequency.

Notably, VIX bridges domains (i) and (iii): when incorporated into the variance equation of GJR-GARCH-X, it substantially improves QLIKE, while simultaneously dominating as a strategy signal. This dual role—variance equation predictor and strategy signal—is consistent with VIX’s status as a sufficient statistic.

A critical methodological caveat: VIX-based backtests suffer from a same-day timing bias of approximately +1.0 Sharpe units when VIX_t is applied to r_t rather than r_{t+1} , because the contemporaneous correlation ($\rho \approx +0.65$) creates a mechanical look-ahead hedge. GARCH-based strategies are naturally immune. All VIX results herein use properly lagged weights.

The complexity ceiling does not imply sophisticated models are useless—they improve risk measurement and regulatory compliance. Rather, it delimits where additional complexity stops improving investment allocation. For daily-frequency portfolios of diversified ETFs, a simple GJR-GARCH forecast combined with inverse-volatility weighting and VIX-implied sizing captures essentially all available information.

We test the VIX sufficient statistic hypothesis across 16 categories of indicators: com-

posite sentiment (CNN Fear & Greed, AAI, Michigan), tail risk (SKEW, VVIX), credit (HY/IG spread, yield curve), macroeconomic (42 FRED series), and Taiwan-specific (27 DGBAS indicators). After controlling for VIX, partial correlations with next-period realized volatility are uniformly below 0.03 for all U.S. indicators. A ridge regression ensemble (leave-one-year-out CV) shrinks all non-VIX coefficients to approximately zero ($\alpha = 100$). A panel GARCH-X with the five strongest candidates *worsens* OOS QLIKE by -8.31% ($p = 0.022$), consistent with overfitting. The sole exception is Taiwan import growth year-over-year ($+5.6\%$, DM $p = 0.043$), consistent with VIX’s incomplete coverage of trade-dependent Asian economies.

Causal discovery (PC algorithm on 12 financial indicators) confirms VIX as an *information sink*: in-degree 4 (receiving edges from credit spreads, equity returns, yield curve, sentiment), out-degree 1. This topology explains why VIX functions as a sufficient statistic: it aggregates causal influences from multiple sources, leaving no residual predictive content after conditioning.

Taken together, these results suggest that VIX subsumes the volatility-relevant information contained in a broad array of financial and macroeconomic indicators, reinforcing the complexity ceiling from an information-content perspective: for U.S. equity volatility, the binding constraint is not model specification but the predictive content available at daily frequency.

F The HAR Paradox: Prediction Superiority Without Economic Improvement

HAR with absolute returns ($|r_t|$) achieves QLIKE of 0.49 versus GJR-GARCH’s 1.51 (DM $= -15.45$, $p < 0.001$).³ This result is universal: HAR-ABS outperforms GJR-GARCH in all seven markets tested (SPY, QQQ, EFA, EWZ, GLD, TLT, 0050.TW), with DM

³The HAR comparison uses the ? centered QLIKE loss, $L^*(\sigma^2, h) = \sigma^2/h - \ln(\sigma^2/h) - 1$, which produces values centered near 1.0. This differs from the quasi-log-likelihood formulation in Eq. (??) used in the main text (values near -9.0). The two formulations are order-preserving: model rankings are identical under both conventions. The centered form is used here because HAR targets $|r_t|$ rather than r_t^2 , making the quasi-log-likelihood scale inapplicable.

statistics from -11.11 to -21.74 , all at $p < 0.001$. The proxy choice is decisive: HAR with r_t^2 produces QLIKE = 1.57, *worse* than GJR—a threefold swing from the same model structure.

Yet HAR-ABS produces the *lowest* VT Sharpe (1.12 vs. 1.75 for 12/VIX) with $1.9\times$ higher turnover and weight correlation of only 0.51 with 12/VIX. The resolution: HAR is backward-looking, optimally aggregating historical magnitudes at daily, weekly, and monthly horizons. VIX is forward-looking, embedding the variance risk premium. For *measurement*, backward aggregation is superior; for *allocation*, forward-looking risk pricing is irreplaceable. This is the fifth independent confirmation of the prediction $\neq$ application paradox in our analysis, sharpening rather than refuting the complexity ceiling.

G Auxiliary Discussions: Crowding Risk and Behavioral Costs

G.1 Crowding Risk

A natural concern is whether widespread adoption of VIX-based VT would erode its effectiveness through crowded trading. We analyze this through three channels. First, the 12/VIX weight function is inherently crowding-resistant: the concave $1/x$ mapping means that coordinated selling during high-VIX episodes produces diminishing marginal impact—each additional dollar of VT selling reduces VIX sensitivity by less than the previous dollar. Second, a feedback-loop simulation in which VT-induced selling raises VIX (which further reduces positions) always converges to a stable equilibrium rather than amplifying. Third, at plausible retail adoption levels ($< \$1\text{B}$ AUM), the market impact is negligible relative to SPY’s daily volume ($\sim \$30\text{B}$). These results suggest VT strategies can be published without responsible-disclosure concerns regarding market stability.

G.2 Behavioral Implications

The asymmetry of behavioral costs for VT users is striking: the cost of abandoning VT during calm markets (FOMO) is approximately $5\times$ larger than the cost of panic-selling during crises. In our simulations, an investor who exits VT after 6 months of underperformance during a bull market and switches to buy-and-hold foregoes the subsequent crisis protection entirely, whereas an investor who panics during a drawdown and sells merely locks in a loss that VT had already substantially reduced. This asymmetry has practical implications for investor communication: VT adoption materials should emphasize the high cost of fair-weather abandonment rather than the (already modest) cost of crisis-period panic.

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